

OS Components & Functions

1. Components of an Operating System

There are two basic components of an Operating System.

Component	Description
Shell	The outermost layer of the Operating System and handles user interaction. It interprets input for the OS and handles the output from the OS.
Kernel	The core component of the operating system. The kernel is the primary interface between the Operating system and Hardware.

2. Applications / Functions of Operating System

Platform for Application Programs

It provides a platform, on top of which, other programs, called application programs can run.

Managing Input-Output Unit

It also allows the computer to manage its own resources such as memory, monitor, keyboard, printer, etc. Management of these resources is required for effective and fair utilization.

Multitasking

It manages memory and allows multiple programs to run in their own space and even communicate with each other through shared memory.

Manages Memory and Files

It manages the computer's main memory and secondary storage. Additionally, it allows and deallocates memory to all tasks and applications.

Provides Security

It helps to maintain the system and applications safe through the authorization process. Thus, the OS provides security to the system.

3. Common Operating Systems

Operating System	Description / Uses
Windows OS	Developed by Microsoft. Used in personal computing, business environment, gaming.
macOS	Developed by Apple. Used for creative industries, personal computing and professional environments.
Linux	Developed by Linus Torvalds and maintained by the open-source community. Used in servers and data centers, development and programming, and personal computing for tech enthusiasts.
Unix	Developed originally at AT&T; Bell Labs. Various commercial and open-source versions are available. Used in servers and workstations, development environments, and research and academic settings.